

# Mathematical Logic

## Definition & Applications

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- ① Learning outcomes
- ② Logic: Definition
- ③ Logic: Simple Applications

# Lecture's Learning Outcomes

Students completing successfully this lecture should be able to...

- 1 **Define** what **Logic** is
- 2 **Describe** **inference** and **proving**.
- 3 **Give** some examples to demonstrate that Logic is able to support inference, reasoning, and proving.
- 4 **Give** some examples to demonstrate the use of Logic in the real world.
- 5 **List** fundamental steps of using Logic to solve a real problem.
- 6 **Identify** some inference rules of Logic

# Logic: Definition

from Hurley [1]

**Logic** may be defined as the science that evaluates arguments.

from Shawn Hedman [2]

A **Logic** is a language equipped with rules for deducing the truth of one sentence from that of another.

# Example 1

for Propositional Logic

## Human's reasoning

**If we have the following facts**

| No. | Satements in English               |
|-----|------------------------------------|
| 1.  | Either Tom or Jerry broke the cup. |
| 2.  | Tom did not break the cup.         |

**then we can **infer** the following fact**

Jerry broke the cup.

Fundamental steps for supporting a formal system, e.g., Computer, to perform the reasoning are presented as the following.

# Example 1

for Propositional Logic

## Computer's reasoning

### STEP 1: Modelling - Symbolizing

| No. | Proposition | English Statement                       |
|-----|-------------|-----------------------------------------|
| 1.  | $T$         | $\triangle \equiv$ Tom broke the cup.   |
| 2.  | $J$         | $\triangle \equiv$ Jerry broke the cup. |
|     | <b>So,</b>  |                                         |
| 3.  | $\neg T$    | means "Tom did not break the cup."      |
| 4.  | $\neg J$    | means "Jerry did not break the cup."    |

# Example 1

for Propositional Logic

## Computer's reasoning

### STEP 1: Modelling - Translating to Propositional Logic

| No.                                                 | English Statement                  | Proposition |
|-----------------------------------------------------|------------------------------------|-------------|
| <b>Translation for premises</b>                     |                                    |             |
| 1.                                                  | Either Tom or Jerry broke the cup. | $T \vee J$  |
| 2.                                                  | Tom did not break the cup.         | $\neg T$    |
| <b>Translation for the statement being inferred</b> |                                    |             |
| 1.                                                  | Jerry broke the cup.               | $J$         |

# Example 1

for Propositional Logic

## Computer's reasoning

### STEP 1: Modelling - The Output

---

**Computer has the following facts**

1.  $T \vee J$
2.  $\neg T$

**Computer wants to infer the following fact**

1.  $J$



# Example 1

for Propositional Logic

## Computer's reasoning

In theory of Propositional Logic, Computer has the following **inference and equivalence rules**.

| No. | Inference Rule              | No. | Equivalence Rule             |
|-----|-----------------------------|-----|------------------------------|
| 1.  | Modus ponens (MP)           | 1.  | DeMorgan (DM)                |
| 2.  | Modus Tollens (MT)          | 2.  | Commutativity (Com)          |
| 3.  | Hypothetical syllogism (HS) | 3.  | Associativity (Assoc)        |
| 4.  | Disjunctive syllogism (DS)  | 4.  | Distribution (Dist)          |
| 5.  | Constructive dilemma (CD)   | 5.  | Double negation (DN)         |
| 6.  | Simplification (Simp)       | 6.  | Transposition (Trans)        |
| 7.  | Conjunction (Conj)          | 7.  | Material implication (Impl)  |
| 8.  | Addition (Add)              | 8.  | Material equivalence (Equiv) |
|     |                             | 9.  | Exportation (Exp)            |
|     |                             | 10. | Modus Tollens (MT)           |

## Disjunctive syllogism (DS)

$$\frac{\neg q, p \vee q}{p} \quad (1)$$

# Example 1

for Propositional Logic

## Computer's reasoning

**STEP 2: Inference** - The reasoning process

| No.                        | Proposition | Note                                                                               |
|----------------------------|-------------|------------------------------------------------------------------------------------|
| 1.                         | $T \vee J$  | Premise (P)                                                                        |
| 2.                         | $\neg T$    | P                                                                                  |
| (The Inference is started) |             |                                                                                    |
| 3.                         | $J$         | 1, 2, DS<br>i.e., Infer Line 3 by applying Inference rule DS on Line 1 and Line 2. |
|                            | □           |                                                                                    |

# Example 1

for Propositional Logic

## Computer's reasoning

### STEP 3: Interpretation

| No.                       | English Statement                  | Proposition |
|---------------------------|------------------------------------|-------------|
| <b>A computer has ...</b> |                                    |             |
| 1.                        | Either Tom or Jerry broke the cup. | $T \vee J$  |
| 2.                        | Tom did not break the cup.         | $\neg T$    |
| <b>It can infer ...</b>   |                                    |             |
| 1.                        | Jerry broke the cup.               | $J$         |

# Example 2

for Propositional Logic

## Human's reasoning

### If we have the following facts

---

No.    Statements in English

---

1.    Tom was in Florida on the new year holiday  
      (i.e., a specific time on the new year holiday).

### We also have implicit rules

1.    If Tom stole the cup in White House on the new year holiday  
      then Tom was in White House on the new year holiday .
2.    On the new year holiday, if Tom was in Florida  
      then Tom was not in White House, and vice versa

### then we can infer the following fact

- 
1.    Tom did not steal the cup.

Fundamental steps for supporting a formal system, e.g., Computer, to perform the reasoning are presented as the following.

# Example 2

## for Propositional Logic

### Computer's reasoning

#### STEP 1: Modelling - Symbolizing

| No. | Proposition | English Statement                                                                |
|-----|-------------|----------------------------------------------------------------------------------|
| 1.  | $T$         | $\triangleq$ Tom stole the cup in White House on the new year holiday.           |
| 2.  | $W$         | $\triangleq$ Tom was in White House on the new year holiday.                     |
| 3.  | $F$         | $\triangleq$ Tom was in Florida on the new year holiday.                         |
|     | <b>So,</b>  |                                                                                  |
| 4.  | $\neg T$    | means "Tom <b>did not</b> steal the cup in White House on the new year holiday." |
| 5.  | $\neg W$    | means "Tom <b>was not</b> in White House on the new year holiday."               |
| 6.  | $\neg F$    | means "Tom <b>was not</b> in Florida on the new year holiday."                   |

# Example 2

for Propositional Logic

## Computer's reasoning

### STEP 1: Modelling - Translating to Propositional Logic

| No.                                                 | English Statement                                                                                                 | Proposition                                            |
|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>Translation for premises</b>                     |                                                                                                                   |                                                        |
| 1.                                                  | Tom was in Florida on the new year holiday.                                                                       | $F$                                                    |
| 2.                                                  | If Tom stole the cup in White House on the new year holiday then Tom was in White House on the new year holiday . | $T \Rightarrow W$                                      |
| 3.                                                  | On the new year holiday, if Tom was in Florida then Tom was not in White House, and vice versa                    | $(F \Rightarrow \neg W) \wedge (W \Rightarrow \neg F)$ |
| <b>Translation for the statement being inferred</b> |                                                                                                                   |                                                        |
| 1.                                                  | Tom did not steal the cup.                                                                                        | $\neg T$                                               |

# Example 2

for Propositional Logic

## Computer's reasoning

### STEP 1: Modelling - The Output

---

**Computer has the following facts and rules**

1.  $F$
2.  $T \Rightarrow W$
3.  $(F \Rightarrow \neg W) \wedge (W \Rightarrow \neg F)$

**Computer wants to infer the following fact**

1.  $\neg T$

## Example 2

for Propositional Logic

Computers have the following **inference and equivalence rules**.

Simplification (Simp):

$$\frac{p \wedge q}{q}$$

Modus Ponens (MP):

$$\frac{p, p \Rightarrow q}{q}$$

Modus Tollens (MT):

$$\frac{\neg q, p \Rightarrow q}{\neg p}$$



# Example 2

for Propositional Logic

## Computer's reasoning

### STEP 2: Inference - The reasoning process

| No.                        | Proposition                                            | Note        |
|----------------------------|--------------------------------------------------------|-------------|
| 1.                         | $F$                                                    | Premise (P) |
| 2.                         | $T \Rightarrow W$                                      | P           |
| 3.                         | $(F \Rightarrow \neg W) \wedge (W \Rightarrow \neg F)$ | P           |
| (The Inference is started) |                                                        |             |
| 4.                         | $(F \Rightarrow \neg W)$                               | 3, Simp     |
| 5.                         | $\neg W$                                               | 1, 4, MP    |
| 6.                         | $\neg T$                                               | 2, 5, MT    |
|                            | $\square$                                              |             |

# Example 2

for Propositional Logic

## Computer's reasoning

### STEP 3: Interpretation

| No.                       | English Statement                                                                                                 | Proposition                                            |
|---------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| <b>A computer has ...</b> |                                                                                                                   |                                                        |
| 1.                        | Tom was in Florida on the new year holiday.                                                                       | $F$                                                    |
| 2.                        | If Tom stole the cup in White House on the new year holiday then Tom was in White House on the new year holiday . | $T \Rightarrow W$                                      |
| 3.                        | On the new year holiday, if Tom was in Florida then Tom was not in White House, and vice versa                    | $(F \Rightarrow \neg W) \wedge (W \Rightarrow \neg F)$ |
| <b>It can infer ...</b>   |                                                                                                                   |                                                        |
| 1.                        | Tom did not steal the cup.                                                                                        | $\neg T$                                               |
| 2.                        | Tom was not in White House.                                                                                       | $\neg W$                                               |

# Example 3

for Predicate Logic

## Human's reasoning

**If we have the following facts**

| No. | Statements in English |
|-----|-----------------------|
| 1.  | Socrates is a man.    |

**We also have implicit rule**

2. All men are mortal.

**then we can infer the following fact**

|    |                     |
|----|---------------------|
| 1. | Socrates is mortal. |
|----|---------------------|

# Example 3

for Predicate Logic

## Computer's reasoning

### STEP 1: Modelling - Symbolizing

| No.        | Predicate               | English Statement                                            |
|------------|-------------------------|--------------------------------------------------------------|
| 1.         | $man(X)$                | $\triangleq$ X is a man, i.e., X is one member of man (set). |
| 2.         | $mortal(X)$             | $\triangleq$ X is mortal, i.e, X is dead-able.               |
| <b>So,</b> |                         |                                                              |
| 3.         | $man(Socrates)$         | means "Socrates is a man".                                   |
| 4.         | $\neg man(Socrates)$    | means "Socrates is <b>not</b> a man".                        |
| 5.         | $mortal(Socrates)$      | means "Socrates is a mortal".                                |
| 6.         | $\neg mortal(Socrates)$ | means "Socrates is <b>not</b> mortal".                       |

# Example 3

for Predicate Logic

## Computer's reasoning

**STEP 1: Modelling** - Translating to Predicate Logic

| No.                                                 | English Statement   | Predicate                                 |
|-----------------------------------------------------|---------------------|-------------------------------------------|
| <b>Translation for premises</b>                     |                     |                                           |
| 1.                                                  | Socrates is a man   | $man(Socrates)$                           |
| 2.                                                  | All men are mortal. | $\forall X(man(X) \Rightarrow mortal(X))$ |
| <b>Translation for the statement being inferred</b> |                     |                                           |
| 1.                                                  | Socrates is mortal. | $mortal(Socrates)$                        |

# Example 3

for Predicate Logic

## Computer's reasoning

### STEP 1: Modelling - The Output

---

**Computer has the following facts and rules**

1.  $man(Socrates)$
2.  $\forall X (man(X) \Rightarrow mortal(X))$

**Computer wants to infer the following fact**

1.  $mortal(Socrates)$

# Example 3

for Predicate Logic

Computer has the following **inference and equivalence rules**.

## Universal instantiation (UI):

Allowed

$$\frac{\forall x F(x)}{F(y)}$$

$F(x)$ : Statement function. Examples:  $[p(x) \Rightarrow q(x)]$ ,  $[p(x) \wedge q(x)]$ , etc

$x, y$ : variables

$a$ : a constant

Allowed:

$$\frac{\forall x F(x)}{F(a)}$$

## Modus ponens (MP):

$$\frac{p, p \Rightarrow q}{q}$$

# Example 3

for Predicate Logic

## Computer's reasoning

### STEP 2: Inference - The reasoning process

| No.                        | Predicate                                    | Note                    |
|----------------------------|----------------------------------------------|-------------------------|
| 1.                         | $man(Socrates)$                              | Premise (P)             |
| 2.                         | $\forall X(man(X) \Rightarrow mortal(X))$    | (P)                     |
| (The Inference is started) |                                              |                         |
| 3.                         | $man(Socrates) \Rightarrow mortal(Socrates)$ | 2, UI, $\{X=Socrates\}$ |
| 4.                         | $mortal(Socrates)$                           | 1, 3, MP                |
|                            | □                                            |                         |



# Example 3

for Predicate Logic

## Computer's reasoning

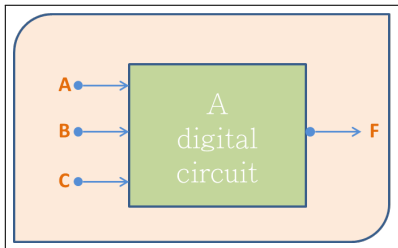
### STEP 3: Interpretation

| No.                       | English Statement   | Predicate                                 |
|---------------------------|---------------------|-------------------------------------------|
| <b>A computer has ...</b> |                     |                                           |
| 1.                        | Socrates is a man   | $man(Socrates)$                           |
| 2.                        | All men are mortal. | $\forall X(man(X) \Rightarrow mortal(X))$ |
| <b>It can infer ...</b>   |                     |                                           |
| 1.                        | Socrates is mortal. | $mortal(Socrates)$                        |

# Example 4

in Digital Logic Design

- **Problem Statement:** Develop a digital circuit having 3 inputs (**A**, **B**, and **C**) and one output (**F**), as shown in the below figure.
- The relationship between the inputs and the output is specified as in the below table



| No. | A | B | C | F |
|-----|---|---|---|---|
| 1.  | 0 | 0 | 0 | 0 |
| 2.  | 0 | 0 | 1 | 0 |
| 3.  | 0 | 1 | 0 | 1 |
| 4.  | 0 | 1 | 1 | 0 |
| 5.  | 1 | 0 | 0 | 1 |
| 6.  | 1 | 0 | 1 | 1 |
| 7.  | 1 | 1 | 0 | 0 |
| 8.  | 1 | 1 | 1 | 0 |

# Example 4

in Digital Logic Design

## STEP 1: Modelling - Symbolizing

| No.        | Proposition | Meaning                                                      |
|------------|-------------|--------------------------------------------------------------|
| 1.         | $A$         | Input $A$ stays on high voltage level, i.e., <b>Level 1</b>  |
| 2.         | $B$         | Input $B$ stays on high voltage level, i.e., <b>Level 1</b>  |
| 3.         | $C$         | Input $C$ stays on high voltage level, i.e., <b>Level 1</b>  |
| 4.         | $F$         | Output $F$ stays on high voltage level, i.e., <b>Level 1</b> |
| <b>So,</b> |             |                                                              |
| 5.         | $\neg A$    | Input $A$ stays on low voltage level, i.e., <b>Level 0</b>   |
| 6.         | $\neg B$    | Input $B$ stays on low voltage level, i.e., <b>Level 0</b>   |
| 7.         | $\neg C$    | Input $C$ stays on low voltage level, i.e., <b>Level 0</b>   |
| 8.         | $\neg F$    | Output $F$ stays on low voltage level, i.e., <b>Level 0</b>  |

# Example 4

in Digital Logic Design

## STEP 1: Modelling - Translating to Propositional Logic

| No. | A | B | C | F |
|-----|---|---|---|---|
| 1.  | F | F | F | F |
| 2.  | F | F | T | F |
| 3.  | F | T | F | T |
| 4.  | F | T | T | F |
| 5.  | T | F | F | T |
| 6.  | T | F | T | T |
| 7.  | T | T | F | F |
| 8.  | T | T | T | F |

# Example 4

in Digital Logic Design

## STEP 2: Designing - Designing of the logic function

| No. | A | B | C | F | Components                                                  |
|-----|---|---|---|---|-------------------------------------------------------------|
| 1.  | F | F | F | F |                                                             |
| 2.  | F | F | T | F |                                                             |
| 3.  | F | T | F | T | $\rightarrow \text{ROW3} = (\neg A \wedge B \wedge \neg C)$ |
| 4.  | F | T | T | F |                                                             |
| 5.  | T | F | F | T | $\rightarrow \text{ROW5} = (A \wedge \neg B \wedge \neg C)$ |
| 6.  | T | F | T | T | $\rightarrow \text{ROW6} = (A \wedge \neg B \wedge C)$      |
| 7.  | T | T | F | F |                                                             |
| 8.  | T | T | T | F |                                                             |

$$\begin{aligned}\text{Function } F(A,B,C) &= \text{ROW3} \vee \text{ROW5} \vee \text{ROW6} \\ &= ((\neg A \wedge B) \wedge \neg C) \vee \\ &\quad ((A \wedge \neg B) \wedge \neg C) \vee \\ &\quad ((A \wedge \neg B) \wedge C)\end{aligned}$$

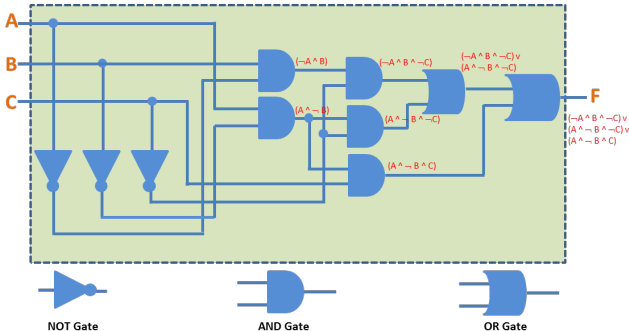
# Example 4

in Digital Logic Design

## STEP 2: Designing - The Output

$$\text{Function } F(A,B,C) = (\neg A \wedge B \wedge \neg C) \vee (A \wedge \neg B \wedge \neg C) \vee (A \wedge \neg B \wedge C)$$

The resulting digital circuit is shown as follows:



# Example 5

in Artificial Intelligence

- **Problem Statement:** Develop a computer program that can play game **MineSweeper** automatically.
- Assume that, it may use a webcam or a camera to detect and recognize the game board, i.e.,
  - It can recognize all of numbers on the game board
  - It can detect which tiles are uncovered and which tiles are empty

# Example 5

in Artificial Intelligence

## MineSweeper's Rules

- There are exactly “N” mines around a tile labelled with number “N”, for examples, “N” = 1,2, and so on.

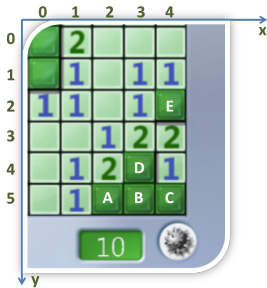


# Example 5

in Artificial Intelligence

## MineSweeper's Example

- Given a game board as shown in the below figure.
- Can you infer the following conclusions?
  - Tiles with Label **A**, **D**, and **E** have mine
  - Tiles with Label **B** and **C** do not have mine



# Example 5

in Artificial Intelligence

## Game's Inputs

- The game program uses a webcam or a camera in combination with computer vision techniques to identify and to recognize numbers ( if available) and classify each cell into one of the three groups as follows: (a) **empty cell**, (b) uncovered **cell**, and **cell with a number**.

## Game's rule

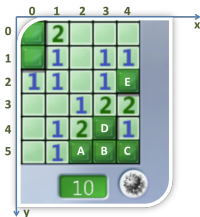
- The game rule has been programmed (for example, as a function) inside the game program. The game program uses the game's rule to generate formal representations as in the following.

# Example 5

in Artificial Intelligence

## STEP 1: Modelling - Symbolizing

| No. | Notation/Proposition | Meaning                                                   |
|-----|----------------------|-----------------------------------------------------------|
| 1.  | $A$                  | There is a mine under Cell $A$ , i.e., Cell (2,5)         |
| 2.  | $B$                  | There is a mine under Cell $B$ , i.e., Cell (3,5)         |
| 3.  | $C$                  | There is a mine under Cell $C$ , i.e., Cell (4,5)         |
| 4.  | $D$                  | There is a mine under Cell $D$ , i.e., Cell (3,4)         |
| 5.  | $E$                  | There is a mine under Cell $E$ , i.e., Cell (4,2)         |
| So, |                      |                                                           |
| 6.  | $\neg A$             | There is <b>no</b> mine under Cell $A$ , i.e., Cell (2,5) |
| 7.  | $\neg B$             | There is <b>no</b> mine under Cell $B$ , i.e., Cell (3,5) |
| 8.  | $\neg C$             | There is <b>no</b> mine under Cell $C$ , i.e., Cell (4,5) |
| 9.  | $\neg D$             | There is <b>no</b> mine under Cell $D$ , i.e., Cell (3,4) |
| 10. | $\neg E$             | There is <b>no</b> mine under Cell $E$ , i.e., Cell (4,2) |



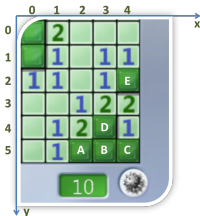
# Example 5

in Artificial Intelligence

## STEP 1: Modelling - Translating to Propositional Logic

for each number on the game board, do the translation

| No. | Location | Value | Variables | Formal Representation                                                                                                                                                          |
|-----|----------|-------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | (1, 5)   | 1     | $A$       | $A$                                                                                                                                                                            |
| 2.  | (1, 4)   | 1     | $A$       | $A$                                                                                                                                                                            |
| 3.  | (2, 4)   | 2     | $A, B, D$ | $(A \vee B \vee D) \wedge (\neg A \vee \neg B \vee \neg D) \wedge$<br>$(\neg A \vee B \vee D) \wedge (A \vee \neg B \vee D) \wedge$<br>$(A \vee B \vee \neg D)$                |
| 4.  | (4, 4)   | 1     | $B, C, D$ | $(B \vee C \vee D) \wedge (\neg B \vee \neg C \vee \neg D) \wedge$<br>$(B \vee \neg C \vee \neg D) \wedge (\neg B \vee C \vee \neg D) \wedge$<br>$(\neg B \vee \neg C \vee D)$ |
| 5.  | (2, 3)   | 1     | $D$       | $D$                                                                                                                                                                            |
| 6.  | (3, 3)   | 2     | $D, E$    | $D \wedge E$                                                                                                                                                                   |
| 7.  | (4, 3)   | 2     | $D, E$    | $D \wedge E$                                                                                                                                                                   |
| 8.  | (3, 2)   | 1     | $E$       | $E$                                                                                                                                                                            |
| 9.  | (3, 1)   | 1     | $E$       | $E$                                                                                                                                                                            |
| 10. | (4, 1)   | 1     | $E$       | $E$                                                                                                                                                                            |



# Example 5

in Artificial Intelligence

## STEP 1: Modelling - The Output

after removing repeated statements

| No. | Statement                                                                                                                                                                      |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.  | $A$                                                                                                                                                                            |
| 2.  | $(A \vee B \vee D) \wedge (\neg A \vee \neg B \vee \neg D) \wedge$<br>$(\neg A \vee B \vee D) \wedge (A \vee \neg B \vee D) \wedge$<br>$(A \vee B \vee \neg D)$                |
| 3.  | $(B \vee C \vee D) \wedge (\neg B \vee \neg C \vee \neg D) \wedge$<br>$(B \vee \neg C \vee \neg D) \wedge (\neg B \vee C \vee \neg D) \wedge$<br>$(\neg B \vee \neg C \vee D)$ |
| 4.  | $D$                                                                                                                                                                            |
| 5.  | $D \wedge E$                                                                                                                                                                   |
| 6.  | $E$                                                                                                                                                                            |

# Example 5

in Artificial Intelligence

## STEP 2: Computer's Reasoning

| No. | Statement                                                                                                                                                                      | Note        |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 1.  | $A$                                                                                                                                                                            | Premise (P) |
| 2.  | $(A \vee B \vee D) \wedge (\neg A \vee \neg B \vee \neg D) \wedge$<br>$(\neg A \vee B \vee D) \wedge (A \vee \neg B \vee D) \wedge$<br>$(A \vee B \vee \neg D)$                | P           |
| 3.  | $(B \vee C \vee D) \wedge (\neg B \vee \neg C \vee \neg D) \wedge$<br>$(B \vee \neg C \vee \neg D) \wedge (\neg B \vee C \vee \neg D) \wedge$<br>$(\neg B \vee \neg C \vee D)$ | P           |
| 4.  | $D$                                                                                                                                                                            | P           |
| 5.  | $D \wedge E$                                                                                                                                                                   | P           |
| 6.  | $E$                                                                                                                                                                            | P           |

The inference is started here

# Example 5

in Artificial Intelligence

## STEP 2: Computer's Reasoning

| No. | Statement                                         | Note                             |
|-----|---------------------------------------------------|----------------------------------|
| 7.  | $(\neg A \vee \neg B \vee \neg D)$                | 2, Simp.                         |
| 8.  | $(\neg A \vee (\neg B \vee \neg D))$              | 7, Assoc.                        |
| 9.  | $A \Rightarrow (\neg B \vee \neg D)$              | 8, Impl.                         |
| 10. | $(\neg B \vee \neg D)$                            | 1, 9, MP.                        |
| 11. | $D \Rightarrow \neg B$                            | 10, Impl.                        |
| 12. | $\neg B$                                          | 11, MP.                          |
| 13. | $B \vee \neg C \vee \neg D$                       | 3, Simp.                         |
| 14. | $\neg(\neg B) \vee \neg C \vee \neg D$            | 13, DN.                          |
| 15. | $\neg(\neg B) \vee (\neg C \vee \neg D)$          | 14, Assoc.                       |
| 16. | $\neg B \Rightarrow (\neg C \vee \neg D)$         | 15, Impl.                        |
| 17. | $(\neg C \vee \neg D)$                            | 12, 16, MP.                      |
| 18. | $(D \Rightarrow \neg C)$                          | 17, Imp.                         |
| 19. | $\neg C$                                          | 4, 18, MP.                       |
| 20. | $A \wedge \neg B \wedge \neg C \wedge D \wedge E$ | 1, 4, 6, 12, 19, Conj. $\square$ |

# Example 5

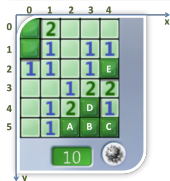
in Artificial Intelligence

## STEP 3: Interpretation of the results

| No. | Statement                                         | Note                             |
|-----|---------------------------------------------------|----------------------------------|
| 20. | $A \wedge \neg B \wedge \neg C \wedge D \wedge E$ | 1, 4, 6, 12, 19, Conj. $\square$ |

### Meaning

1. There is a mine under Cell A, i.e, (2,5)
2. There is **no** mine under Cell B, i.e, (3,5)
3. There is **no** mine under Cell C, i.e, (4,5)
4. There is a mine under Cell D, i.e, (3,4)
5. There is a mine under Cell E, i.e, (4,2)





In this lecture, learners have studied the following

- The definition of Logic.
- Some examples of using Propositional Logic and Predicate Logic.
- Fundamental steps for solving a real problem using Logic.
- The definition and the usage of reasoning and proving.
- Definition of some concepts as follows: inference rules, equivalence rules, and facts.
- The name and the way of using several inference rules.

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